

Derivation of the Distribution of $S(t)$ Compound Poisson with Exponential Jumps

Problem setup

Let $N(t)$ be a Poisson process with rate $\lambda > 0$ (equivalently exponential interarrival times with parameter λ). Let the jump sizes X_i be i.i.d. $\text{Exp}(\mu)$ (rate $\mu > 0$), independent of N . Define the compound process

$$S(t) = \sum_{i=1}^{N(t)} X_i,$$

with the usual convention that an empty sum (when $N(t) = 0$) equals 0.

We derive the law of $S(t)$: it has an atom at zero and a continuous density on $(0, \infty)$ expressible in closed form.

1. Laplace transform (MGF) approach

Condition on $N(t)$. For an individual exponential jump $X \sim \text{Exp}(\mu)$, the moment-generating function (for $u < \mu$) is

$$M_X(u) = E[e^{uX}] = \frac{\mu}{\mu - u}.$$

Conditioning on $N(t)$ gives

$$M_{S(t)}(u) = E[e^{uS(t)}] = E[E(e^{uS(t)} \mid N(t))] = E[M_X(u)^{N(t)}].$$

Since $N(t) \sim \text{Poisson}(\lambda t)$ with pgf $E[s^{N(t)}] = \exp(\lambda t(s - 1))$, we obtain

$$M_{S(t)}(u) = \exp\{\lambda t(M_X(u) - 1)\} = \exp\left\{\lambda t\left(\frac{\mu}{\mu - u} - 1\right)\right\}.$$

For the Laplace transform (set $u = -s$, $s > 0$):

$$\mathcal{L}\{f_S(\cdot)\}(s) = E[e^{-sS(t)}] = \exp\left\{\lambda t\left(\frac{\mu}{\mu + s} - 1\right)\right\}.$$

2. Atom at zero

The probability of no jumps by time t is

$$\Pr\{S(t) = 0\} = \Pr\{N(t) = 0\} = e^{-\lambda t}.$$

Therefore the distribution has an atom of mass $e^{-\lambda t}$ at 0.

3. Continuous part for $s > 0$ (mixture representation)

Condition on $N(t) = n \geq 1$. The sum of n independent $\text{Exp}(\mu)$ variables has an Erlang (Gamma) density:

$$f_{S|N=n}(s) = \frac{\mu(\mu s)^{n-1} e^{-\mu s}}{(n-1)!}, \quad s > 0.$$

Unconditioning (mixture over the Poisson counts) yields for $s > 0$:

$$\begin{aligned}
f_S(s) &= \sum_{n=1}^{\infty} \Pr\{N(t) = n\} f_{S|N=n}(s) \\
&= \sum_{n=1}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} \cdot \frac{\mu(\mu s)^{n-1} e^{-\mu s}}{(n-1)!} \\
&= e^{-\lambda t - \mu s} \mu \sum_{n=1}^{\infty} \frac{(\lambda t)^n}{n!} \frac{(\mu s)^{n-1}}{(n-1)!}.
\end{aligned}$$

4. Summation and Bessel function identity

Shift the index $k = n - 1$ (so $k \geq 0$), and factor constants:

$$\begin{aligned}
f_S(s) &= e^{-\lambda t - \mu s} \mu \sum_{k=0}^{\infty} \frac{(\lambda t)^{k+1}}{(k+1)!} \frac{(\mu s)^k}{k!} \\
&= e^{-\lambda t - \mu s} \sqrt{\frac{\lambda \mu t}{s}} \sum_{k=0}^{\infty} \frac{(\lambda \mu t s)^{k+\frac{1}{2}}}{k! (k+1)!}.
\end{aligned}$$

Use the modified Bessel function of the first kind (order 1) identity:

$$I_1(2z) = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{k! (k+1)!}, \quad z \in \mathbb{R}.$$

Set $z = \sqrt{\lambda \mu t s}$. Then

$$\sum_{k=0}^{\infty} \frac{(\lambda \mu t s)^{k+\frac{1}{2}}}{k! (k+1)!} = I_1(2\sqrt{\lambda \mu t s}).$$

Therefore the continuous part of the density for $s > 0$ is

$$f_S(s) = \exp(-\lambda t - \mu s) \sqrt{\frac{\lambda \mu t}{s}} I_1(2\sqrt{\lambda \mu t s}), \quad s > 0.$$

Combined with the atom at zero, the full law is

$$\Pr\{S(t) = 0\} = e^{-\lambda t}, \quad f_S(s) \text{ (} s > 0 \text{) as above.}$$

5. Remarks and asymptotics

- The distribution is a classical compound Poisson: an atom at zero (no jumps) plus an absolutely continuous mixture of Erlang densities.
- Parameter effects:
 - Increasing λ increases the expected number of jumps $E[N(t)] = \lambda t$, reduces the mass at zero, and tends to shift the distribution to larger values (more jumps).
 - Increasing μ reduces the mean jump size $1/\mu$, concentrating the continuous density nearer to 0 and reducing tail mass.
- For large λt (many jumps), the atom at zero becomes negligible. By the central limit theorem the sum of many i.i.d. jumps is approximately normal in the central region, with

$$\mathbb{E}[S(t)] = \mathbb{E}[N(t)]\mathbb{E}[X] = \frac{\lambda t}{\mu}, \quad \text{Var}(S(t)) = \lambda t \text{Var}(X) = \frac{\lambda t}{\mu^2}.$$

Nevertheless the exact distribution remains positively supported and skewed; the Bessel-form density above gives the precise shape and tail decay.

6. Useful formulae summary

$$\Pr\{S(t) = 0\} = e^{-\lambda t}, \quad f_S(s) = e^{-\lambda t - \mu s} \sqrt{\frac{\lambda \mu t}{s}} I_1(2\sqrt{\lambda \mu t s}), \quad s > 0.$$

Reference: The form with modified Bessel I_1 arises from summing the Poisson-weighted Erlang densities and applying the Bessel-series identity.